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Improving interdisciplinary teamwork competence and team functioning in higher education using a role-based reflective teamwork improvement method

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Introduction: Interdisciplinary research, often done in teams, is needed to face complex problems. This means that teamwork competence is becoming ever more important in researchers. Educators in higher education have incorporated teamwork learning in their programs, but what is often lacking is explicit teamwork competence teaching and learning. Teaching teamwork is complicated, as there are about 94 teamwork competence areas in which a person and/or team requires competence for good teamwork and about 250 different aspects of teamwork can be differentiated. The aim of this paper is to provide a developed and exploratively tested method for explicitly teaching teamwork competences in interdisciplinary higher education programs.

Methods: The proposed method, the Role-based Reflective Teamwork Improvement Method (RRTIM), builds on four roles that students take on during their teamwork. These roles are in line with overarching aspects of team work: Task Manager, Process Manager, Relationship Manager and Communications Manager. Using an educational research design path, the method was piloted and improved based on feedback from both students and teachers. The final version of the RRTIM was then evaluated by comparing similar reflections in a non-RRTIM course and a course in which the RRTIM was implemented.

Results and discussion: Exploratory findings suggest that in the course where the RRTIM was implemented, students showed greater awareness of different aspects of teamwork, slightly more in-depth and nuanced reflections upon and analysis of their own teamwork, and more indications of agency in changing sub-optimal team processes and strengthening already good team processes.

KEYWORDS

higher education, interdisciplinary teamwork, reflective learning, role-based learning, teamwork competence

Introduction

As Riebe et al. (2016) state, teamwork is no longer just a desirable competence, but an essential competence. To face complex problems, interdisciplinary research is undertaken, in which different disciplinary perspectives come together (Repko and Szostak, 2021). This type of research is done in teams, bringing together team members with different disciplinary backgrounds to study a complex problem (Repko and Szostak, 2021).

However, effective interdisciplinary teamwork requires competence, as it is complex, highly dynamic and unpredictable in itself (Kozlowski and Ilgen, 2006; MacLeod, 2018).

Educators in higher education have incorporated teamwork learning in their programs, but what is often lacking is explicit teamwork competence teaching and learning (Earnest et al., 2017; Hansen, 2006). Either there is an assumption that students will learn teamwork competence from just exposure to teamwork alone, or there is not sufficient time or expertise to teach teamwork competence in already cramped programs (van Woerden et al., 2025a).

Teaching teamwork is complicated, as there are about 94 teamwork competence areas in which a person and/or team requires competence for good teamwork and about 250 different aspects of teamwork can be differentiated (van Woerden, 2025). Interdisciplinary teamwork poses extra challenges, as these teams are characterized by high disciplinary diversity, which can lead to misunderstanding, deep disagreement and conflict, due to a lack of shared mental models (van Woerden et al., 2025a; Jehn et al., 1999; MacLeod, 2018; Miller and Mansilla, 2004). Programs are struggling to find ways to teach these complicated competences, in limited time.

The aim of this paper is to provide a developed and exploratively tested method for explicitly teaching teamwork competences in interdisciplinary programs. Students have indicated that they need more explicit teaching in how to manage interdisciplinary teamwork, that they want to reflect on their own teamwork, and that they would like to learn different team roles with the related competences (van Woerden et al., 2025a). In this paper we aim to design a teamwork teaching and learning method that takes into account these student needs. With this purpose in mind the following research question was formulated: how can we design and implement a role-based reflective method for teaching teamwork competence in interdisciplinary programs?

Teamwork

Teamwork in general consists of about 250 related aspects which are often categorized in input-process-output models of teamwork (e.g., van Woerden, 2025; Tannenbaum et al., 1992). These aspects of teamwork are all interrelated, and can be dynamic and emergent in nature (Kozlowski and Ilgen, 2006). *Inputs* exist before the team starts to work together and consist of (a) individual characteristics (e.g., competences), (b) individual characteristics in relation to a specific team or task (e.g., motivation for the task), (c) task characteristics (e.g., the complexity of the task), (d) team characteristics (e.g., the diversity in the team), and (e) work structures (e.g., procedures and rules; van Woerden, 2023; Tannenbaum et al., 1992). The *process* starts when the team actually starts to work on the task together. Here a distinction can be made between the team processes (how the team acts) and team emergent states (the cognitive, motivational, and affective states of the team; Marks et al., 2001). Examples of team processes are communication, performance monitoring, conflict management, planning and coordination (Kozlowski and Ilgen, 2006; Marks et al., 2001). Team emergent states such as cohesion and trust usually have positive effects on teamwork (Costa et al., 2018; Kozlowski and Ilgen, 2006), whereas conflict (especially relationship conflict)

may be detrimental for teamwork and team performance (Jehn et al., 2008). Finally, the *output* of teamwork is both the task output (e.g., the product) and the team output (e.g., team viability, team member satisfaction).

Interdisciplinary teamwork

Interdisciplinary teamwork is a specific kind of teamwork, which is characterized by high task complexity, high task uncertainty, high (disciplinary) diversity, and high task interdependence (van Woerden, 2025; Fiore, 2008; Repko and Szostak, 2021). Interdisciplinary research is undertaken to analyse or solve complex problems. These complex or 'wicked' problems are usually ill-defined, have many interrelated factors and actors and change constantly. The task of studying these complex problems is therefore quite complex, and requires challenging interdisciplinary and imaginative thinking from the team members (Cantor et al., 2015). The task is also highly uncertain, as team members cannot rely on reified disciplinary methods of doing research, but have to develop their own methods for integrating knowledge from different disciplines (Repko and Szostak, 2021). Complex problems are most often studied by teams with members with different disciplinary backgrounds, which means they use different theories, methods, and concepts, but also that they may have very different assumptions about science or the phenomenon that they study (MacLeod, 2018; Repko and Szostak, 2021). These differences may lead to deep disagreement or misunderstanding, which can severely hamper the teamwork (van Woerden et al., 2025a; MacLeod, 2018). Because all team members represent a body of knowledge or a discipline, team members are highly dependent on each other, as they cannot do each other's work (Fiore, 2008). Positive interdependence theory states that high interdependence can positively influence team performance and can help avoid conflict (Johnson and Johnson, 2009; e.g., Lee et al., 2015). However, high task interdependence can also lead to reduced individual autonomy (Langfred, 2005) and it may lead to high frustration levels, if one team member fails to contribute and exhibits free-riding behaviours, which is common in higher education settings (Hall and Buzwell, 2013).

Student interdisciplinary teamwork

Student teams have their own characteristics. Many student teams do not work together for more than one course, and therefore student teams do not have incentive to dedicate much time to interpersonal relationship building, cohesion and trust building, or to work on team competence development (Bacon et al., 1999; Burdett, 2003; Hansen, 2006; Pfaff and Huddleston, 2003). Free-riding, social loafing and unequal workload division are much reported problems in student teams. Most student teams are outcome dependent in the sense that they receive a team grade for their efforts. Thus, individual students can decide to contribute less and benefit from the work of others. However, many free-riding behaviours may also be attributed to the fact that students do not always feel confident or competent enough to contribute (Hall and Buzwell, 2013). Interdisciplinary

tasks are complex and students in undergraduate programs may lack disciplinary expertise, the competences to understand the work of other disciplines and they are not yet experts at interdisciplinary research methods (van Woerden et al., 2025b; Horn et al., 2023). Lack of expertise may lead to personal feelings of inadequacy or shallow discussions (Kiernan et al., 2020). In addition, most programs do not teach students teamwork competence explicitly. Educators are not always able, competent or willing to scaffold and guide teamwork processes (Riebe et al., 2016), so students lack the skills to address suboptimal team processes, frustrations, and conflict (Abbasi et al., 2017; Fiore et al., 2019; Hansen, 2006).

Learning interdisciplinary teamwork

Student interdisciplinary teamwork is difficult to manage, because some of the characteristics make it more challenging than other types of teamwork. Students have indicated that they need more explicit teaching in how to manage interdisciplinary teamwork, that they need time to learn, practice, and reflect on interdisciplinary teamwork competences, and that they would like to learn different team roles with the related competences (van Woerden et al., 2025a).

Existing teamwork teaching methods, such as the Team CARE model, have focused on assessing team functioning with a tool (O'Neill et al., 2018), but students have indicated they also would like to have more agency and learn how to discuss and to change sub-optimal team processes (van Woerden et al., 2025a). Role-based learning has been shown to be an effective approach in cooperative learning as well as in higher education (Loh and Ang, 2020; Wills et al., 2011), and could potentially provide that agency to students. Cooperative learning methods have included roles (such as facilitator, recorder, presenter, resource manager) to structure teamwork learning, and to make students more active in their learning (Johnson et al., 1994).

According to Forsyth (2014) roles are a natural element in teams. It is very common that team members take up roles within a team (e.g., the role of leader or coordinator), either because they develop in interaction, because they are deeply ingrained in a person or because they have a functional utility to the team. These self-assigned or pre-assigned roles define their responsibilities and the expectations others may have of them. Organizing the cooperative learning roles (and associated competences) into higher categories, Forsyth (2014) distinguishes two roles based on the core aspects of teamwork. Task roles, which are geared towards the task of the team, are conceptually distinguished from relationship roles, which help satisfy the interpersonal and emotional needs of team members (Forsyth, 2014). Within the task role, we conceptualize a further distinction between roles that are focusing on the task itself (*task execution roles*) and roles that focus on organizing the process for task execution (*process roles*) such as planning, task division, and monitoring of progress. We also further distinguish between *relationship roles*, which are more concerned with the interpersonal processes in the team and *communication roles*, which are concerned with the quality of communication. Each of these roles requires a different set of behaviours and competences (Forsyth, 2014), which can be

learned and practiced. Most people have a certain preference for a specific role, but could also benefit from learning other roles (and the related competences) in teamwork. Using assigned roles in student interdisciplinary teamwork has two potential benefits. First of all, by using assigned roles in which all responsibilities in the team are covered, team functioning could be improved because no aspect of teamwork will be left unaddressed if roles are executed well. Secondly, by learning different roles, students have the opportunity to learn competences that may be out of their natural comfort zone and skill set, as they may be assigned roles that do not naturally suit them. The four types of roles (task execution, process, relationship, and communication roles) have been used as a base for the design of the role-based team reflection intervention. Some similar interventions with a division in roles between cognitive leader, metacognitive leader and socio-emotional leader have shown moderately positive results, especially of the effect of the role of socio-emotional leader (Cheng et al., 2014).

Combining both reflection on interdisciplinary teamwork with defined teamwork roles may be helpful for students to gain teamwork competence and improve their team functioning. Reflection is described as a powerful tool for deep learning (Kolb, 1984; Moon, 2004), and for learning in complex and dynamic environments (Schön, 1987). According to Schön (1987), expertise in any field is built from practice through reflection on action (after the experience), as well as reflection in action (during the experience). Interdisciplinary teamwork is complex and dynamic, and requires team members to identify and change sub-optimal team processes during the teamwork (Kozłowski and Ilgen, 2006), which can be done through consecutive cycles of reflection. Reflection competence is therefore conceptualized as part of teamwork competence. In higher educational settings, reflective practice is used to help students reach higher levels of understanding (Moon, 2001).

Reflection on their own competences, their teamwork processes and the (disciplinary) differences between team members may help students understand and evaluate their own teamwork (van Woerden et al., 2025a). There is some empirical support for the positive effect of reflection on teamwork competence, the quality of teamwork processes and team performance although these findings are exploratory and some studies do not report positive effects (Horn et al., 2022; Moreland and McMinn, 2010; Wiedow and Konradt, 2011). Reflection could help increase teamwork competence, although most research focuses on reflection as an assessment method of teamwork competence (e.g., Bosman et al., 2020; e.g., Farreras Esclusa and Bofill Soliguer, 2022). Teamwork process improvement in the form of reflection on team processes has been theorized to improve team outputs through improvement of the team processes themselves and there is some empirical evidence to substantiate that claim (Wiedow and Konradt, 2011). O'Neill et al. (2017) concluded that reflexivity training has a positive effect on conflict experience in student teams. Also, team reflection can improve the quality of team emergent states such as team shared mental models (Konradt et al., 2015).

Hoyrup and Elkjaer (2006) have tried to categorize current thinking and theories on reflection in three categories or perspectives, as they call it: the individual perspective, the social-relations perspective and the organizational perspective. The individual perspective mainly focusses on individual learning based on individual reflection. The social-relations perspective

focusses on individual as well as group learning in reflective dialogues. The organizational perspective focusses on sharing perspectives to enhance work situations and change work structures. The teaching and learning method we developed combines the individual and the social-relations reflective practice.

Materials and methods

Intervention design

Based on the literature review and our experience as teachers and program managers, we designed the Role-based Reflective Teamwork Improvement Method (RRTIM). Using an educational design research-based approach (McKenney and Reeves, 2019), the intervention was developed and piloted (PART 1), redesigned (PART 2) and evaluated (PART 3) in three iterative design cycles. The method redesigned in PART 2 consists of different elements to teach students teamwork competence: (a) a lecture on team dynamics, (b) a team charter exercise, (c) four differentiated roles, a manual for each role (see [Addendum II-VI](#)), training in the roles for each student, (d) team conversations to discuss team dynamics based on the observations of each student from their role (with a prior short lecture on individual and social-relational reflection), and to reflect on the team charter, and (e) a final reflection on the team processes and dynamics. The teachers received a teacher training to guide the students using the RRTIM. The differentiated roles are those of Task Manager, Process Manager, Relationship Manager and Communications Manager. The Task Manager is in charge of overseeing the quality of the execution of the task and the end product, which in interdisciplinary research environments means benefitting from different perspectives while at the same time bridging disciplinary differences by first diverging and then converging knowledge. The Process Manager is in charge of overseeing the quality of the process of completing the task, including planning, task division, document management and progress monitoring. The Relationship Manager is in charge of overseeing the quality of the teams relationships, as well as the affective state of team members, by monitoring affect, frustration, relationship conflict and interpersonal relationship building. The Communications Manager is in charge of overseeing the quality of the communication, both internally (team communications channels, the content and modes of communication) as externally (communication with the teacher or other stakeholders). The method can be used in a minimal scaffolding version, for example at the beginning of the learning trajectory, in which student not yet use the roles or have team conversations based on the roles, but do draw up a team charter and reflect on the team charter and do a final reflection on the team dynamics. Later in the program, when students are used to the team charters and reflection thereon, the full scaffolding of the roles and team reflective conversations can be introduced.

Data collection

In PART 1 (2019–2021), the first version of the RRTIM was developed and pilots were conducted in three different programs, an interdisciplinary sustainability program (program 1), an interdisciplinary natural and social sciences program

(program 2), and an interdisciplinary social sciences program (program 3). Thirty students (third year interdisciplinary thesis course, program 1), 120 students (first year introduction into interdisciplinary research course, program 2), 100 students (third year interdisciplinary project course, program 2), and 90 students (first year inter- and transdisciplinary project course, program 3) participated in the researched courses. We deliberately included students from different years in the designing phase of the study, to make sure that the RRTIM was useful for students in all phases of their bachelor education. In each of these pilots, different data was collected to evaluate and redesign the course (see [Table 1](#)). All participants in the research consented to the collection of their data.

In PART 2 (2021–2025), the RRTIM was redesigned based on the feedback in PART 1 and continuously developed over the course of four academic years based on formal student course evaluations, informal conversations with teachers, and informal interviews with experts both within academia as well as in other workfields (for example teamwork trainers and representatives from Deep Democracy). In PART 3 (2025–2026), we were interested in the differences between minimal scaffolding and the full implementation of the RRTIM to see if students improved their level of reflection on teamwork and their teamwork competence. This was analysed in program 3's first year inter- and transdisciplinary project courses, by comparing course 1 (minimal scaffolding of teamwork) and course 2 (full RRTIM implemented). In course 1, 233 students participated, of which 65 consented to participate in the research. In course 2, 230 students participated, of which 137 agreed to participate in the research. Consent was asked in a different context (during the break of a final presentation event vs. a normal working group class), which may explain the difference in consent rate, but of course self-selection effects cannot be ruled out. Between the two courses, there was an overlap of 41 students who agreed to participate in the research in both courses. Since our aim was also to exploratively compare minimal scaffolding to full RRTIM implementation, we decided to research this group of 41

TABLE 1 Overview of data collection.

Course	Data collected (PART 1)
Third year thesis course, program 1	researcher notes
Third year interdisciplinary project course, program 2	Teacher interviews, student written feedback, researcher notes
First year introduction to interdisciplinary research course, program 2	Student focus group, researcher notes
First year (2nd semester) inter- and transdisciplinary project course, program 3	researcher notes (also based on informal talks to students and teachers)
	Data collected (PART 3)
First year (1st semester) inter- and transdisciplinary project course, program 3	Team reflection assignment (students receive minimal scaffolding for teamwork)
First year (2nd semester) inter- and transdisciplinary project course, program 3	Team reflection assignment (students have been introduced to the full RRTIM)

students. Unfortunately, three students did not hand in assignments in both courses, so 38 students remained for analysis. Of these students, their final reflection was analysed. In both final reflections students were asked identical questions (a) 'What went well in your collaboration? How did you make use of that? Give at least one concrete example.' (b) 'What went less well in your collaboration? What did you do to change that? Did it work? What would you have liked to do differently? Give at least one concrete example.' (c) 'What is the most important thing you have learned about collaboration during this course? Explain and give an example.' In course 2 an extra question was added about the RRTIM roles: 'You had a specific role within this collaboration (such as task manager, process manager, interpersonal relations manager and communications manager). How were you able to fulfill this role? Which elements of that role were executed well? Which less so?'

The study protocol for PART 3 was approved by the Ethics Advisory Board of the Amsterdam Institute for Social Science Research of the University of Amsterdam, with reference number: FMG-11858.

Data analysis

In PART 1 we were mainly interested in the usefulness of the method and in what could be improved in the method. Therefore, data analysis of the focus groups and interviews was based on open coding within three pre-defined themes of 'what is useful in the RRTIM', 'what is not useful in the RRTIM', 'what can be improved in the RRTIM'. Then axial coding was used to further cluster the codes within themes. For PART 3, we were interested to see if there were differences between the minimal teamwork scaffolding course and the full RRTIM implementation course in students' level of teamwork reflection and teamwork competence. To make the comparison, data analysis of the team reflection assignments was done using a template coding for (a) level of reflection of teamwork, (b) the (sub-)optimal teamwork processes students had identified in their team, (c) the activities students had undertaken to improve their teamwork, (d) the activities students would have liked to undertake, but did not. The level of reflection on teamwork was scored in a rubric, based on the work of Moon (Moon, 2001) and Grossmann (Grossman, 2009), the score sheet can be found in [Addendum I](#). In the analysis the level of reflection scores for the same student were compared between the two courses. The (sub-)optimal teamwork processes students had identified in their team were template coded based on the comprehensive overview of interdisciplinary teamwork aspects of [van Woerden \(2023\)](#). In the analysis the number of teamwork aspects that the same student reflected on in the different courses was compared, to see if students had widened their scope of teamwork aspects to reflect upon. The activities students had undertaken or would have liked to have undertaken to improve their teamwork were open coded and the former were later organized in overarching categories of 'individual' and 'team' actions. The scoring rubric for reflection and the teamwork processes framework were discussed in the research team before using them. In earlier research the researchers had calibrated on using the teamwork aspects template coding scheme. The final scoring/coding was

done by the first author. The quotes used in this paper were translated by the first author from Dutch to English.

Positionality of the authors

The authors have much experience with interdisciplinary student teamwork and during their years of teaching have had informal conversations with students about their teamwork learning needs. The first author was very interested in the phenomenon that students have many needs when it comes to learning interdisciplinary teamwork competences which are not catered for, and completed a PhD project to further research the phenomenon. As a side project in this PhD, which was mainly aimed at analyzing team dynamics in interdisciplinary student teams, the first author felt the need to also change current teamwork teaching practices, and looked for collaborators to design a teamwork teaching method. The authors have developed the RRTIM method together, and are therefore convinced of its potential use, and invested in its further development. This could be seen as a conflict of interest, but our main aim is to share the progress we have made and to invite others to help us improve the method to further cater to student needs. As teamwork is dynamic, emergent and always changing, we do not believe there is a one-size fits all solution to all teamwork problems, but we do want to put effort into helping students with their teamwork. We believe the RRTIM can do this, based on our data analysis.

Results

Validation of the role-based reflection intervention PART 1: design and piloting of the intervention

In the first pilot in a third year thesis course in program 1, the use of the method was voluntary. Based on researcher notes, we analyzed that students who did partake in the pilot seemed to benefit from the method, as they learned how to discuss and improve their teamwork processes. However, teachers indicated that most of the participating teams were already 'good teams'. The teams that experienced serious issues did not join the pilot as they perceived it as a waste of time. Another key observation was that students needed more guidance in team work, and also in take on the roles of the manuals. Just a manual was too little embedding in the course for the students to actually work with the roles and the manuals.

In the second pilot in a third year interdisciplinary project course in program 2, the three teachers (T1, T2, T3) commented that the RRTIM worked well, both for groups that were functioning well (T1) as for teams that were not functioning well (T3), and that they wanted the RRTIM to be used in the future. Students took their roles seriously (T1) and were enthusiastic about the roles (T3), although their engagement became less as the course progressed (T1 and T3). T2 liked how they gained more insight into how students collaborate, and that they could guide the collaboration where necessary. The teachers saw especially good improvements for the Process Management and the Communication Management

(T1 and T2), whereas Task Management and Relations Management seemed to be more difficult for students (T1 and T2). Overall they were enthusiastic about the RRTIM, but they also had some recommendations on how to improve the RRTIM.

First of all, the teachers indicated that they did not feel fully prepared to guide students in their collaboration and would have liked to have additional training. T1 added that they are much more equipped as research supervisor than as collaboration supervisor and that students sense that their teacher is undertrained in giving collaboration guidance. T3 indicated that they had too little time themselves to really deepen their knowledge of team work. Secondly, the teachers made comments about the structure of the RRTIM itself. They found the manuals and assignments useful and giving structure (T1 and T3), but the structure could be improved even more (T1), the manuals were quite long (T3), and the assignments quite big (T3). Also the students needed more guidance with the Relations and Task Management role (T2), which could also be deepened (T1). Relations are hard for students to manage, so the focus on Relations became less during the course (T2). Also, the boundaries between roles were not always clear (T1). Finally, the teachers made comments about the embedding of the RRTIM in the course. They felt that the focus on collaboration, the engagement with the roles and the use of the manuals became less in the second part of the course (T1 and T3). One of the reasons is that the teacher has too little time to guide the students in the collaboration (T3). Teachers need more time during the course to pay attention to the RRTIM (T3). T3 also mentioned that students need to learn reflection skills before or during the course in order to be able to carry out the role effectively. With regard to assessment, T1 added that the collaboration process should be graded separately from the research product.

All the students of the third year interdisciplinary project course in program 2 were asked to fill out a short evaluation form during one of the lectures in which the roles were explained in further detail. They were asked to provide tips and tops about the RRTIM in general, and the given lecture in specific. 41 students handed in their evaluation. Table 2 provides an overview of students' feedback. Students brought forward 123 unique comments, which can be clustered in themes.

What could be taken from this overview is that overall students thought explicit focus on team work is important, and a role-based method works well. However, they also had useful comments for further embedding of the RRTIM into the course, the guidance provided by teachers, the elements of the RRTIM itself and they highlighted problems of execution.

Finally, a student focus group was organized with four students, of which two participated in the first year introduction to interdisciplinary research course in program 2 (F1 and F2), one in the third year interdisciplinary project course in program 2 (F3), and one was a former student member of the study program committee who had taken both courses before the changes (F4). The purpose of the focus group was to understand what students needed to enhance their teamwork skills and how to incorporate that into the RRTIM in PART 2.

Students indicated that they thought it was very useful to learn how to address issues in their teams and they had experienced and heard many stories of how collaboration in student teams can go wrong. As student F4 put it:

The challenge in collaboration is much more about addressing the group process, the expectations and the responsibilities [in the group]. And that is very difficult, because you don't know each other or you are collaborating with friends, so you do not want to get into conflict. This is hard, and gets much less attention than models of collaboration or feedback. (...) For me the biggest difficulty in collaboration is to create a group culture in which everyone feels free to give their opinion and also to voice what others do wrong or what they do right (...) it can be very confrontational and difficult [to discuss issues in the group], but that is exactly where the major learning objective lies for me when it comes to collaboration.

However, students did report some issues with the current form of the RRTIM. They indicated that they at times thought the roles were confusing, because it was unclear whether students should do all the tasks that belonged to the role, or just see to it that the tasks were executed (F3). Both F2 and F3 reported that students did not stick to their role, however F2 added that the role structure did make clear to the students that there were more aspects to collaboration than they thought originally, and F1 added that the focus on shared and assigned responsibility worked well in the team. F2, F3 and F4 advocated a more free form in which tasks were not assigned to one person through the roles, but were all seen as team tasks. With regards to assignments, F1, F3 and F4 advocated a format with less rigid assignments and a more free format in which students are encouraged to discuss their group process and reach agreements about their own process, without a written reflection requirement.

As researchers we also gathered some additional reflections on the RRTIM, in addition to the above remarks of both students and staff. What became clear from teacher reactions was that the extra time investment of a new training and guidance of students in collaboration was seen as an extra burden, on top of their already high work load. The teachers saw the importance of the RRTIM, but did not feel they had enough time to internalize the method and did not feel equipped to coach students in their collaboration, so they often gave less attention to the RRTIM than was asked of them, which seemed to hamper the implementation and positive impact of the method. The teachers also showed a clear preference for ready-to-use materials, that could be incorporated easily in a class, without too much preparation on their side. Finally, the reading structure of the manuals seemed to be less effective than intended, because the how-to-fix-problem-X structure caused some redundancy and repetition and researchers got the impression that students stopped reading at a certain point.

Validation of the RRTIM PART 2: redesign of the RRTIM

The RRTIM was changed based on the evaluation in PART 1 in seven ways: 1) from the start, the training was embedded in the course and introduced to the teachers and coordinator of the course, 2) the teachers were offered a training session to explain the RRTIM and to make them more comfortable with

TABLE 2 Overview of student' comments on the RRTIM.

Theme	unique comments	students
What is useful in the RRTIM		
Assessment of the RRTIM as important/useful/good/informative	20	19
Positive assessment of the division of tasks and responsibilities into the four roles, such as: (1) the role division works well, it is clear which tasks can be identified in collaboration and the tasks are executed (2) students thought it useful to learn how to manage team dynamics through the roles (3) the role gives students legitimacy and freedom to address issues and to increase accountability (4) students learned to look at their collaboration in a new way	16	11
Positive comments about being able to discuss problems or issues with collaboration in the team as well as the class	7	6
Positive comments about didactic choices made in the lecture	10	7
Positive outcomes of the RRTIM, such as: (1) students took the project more serious (2) it saved time that everyone was aware of the different tasks beforehand (3) students started on the project more structured (4) students became aware of how group dynamics work and how the dynamic may be improved (5) students appreciated the extra help when they had issues in their team (6) students became more aware of their own contribution to collaboration (7) students gained more overview over their project (7) students gained more insight into collaboration (9) students could directly apply what they had learned (10) students became interested and wanted to learn more (11) students learned to look for what motivated team members (12) students learned to record important agreements in writing	13	10
What can be improved in the RRTIM		
Make more clear what the value, the use, the relevance and the importance of team work training is, also for well-functioning teams	7	10
Embed the RRTIM more into the course, for example: (1) provide training in the roles right at the beginning of the course (2) integrate the RRTIM more into the course, as it now feels separate and introduced quite late (3) balance the workload as the RRTIM increases the workload in an already demanding course (4) make the learning objective for the RRTIM part of the course more clear (5) make a separate course for learning the RRTIM (6) embed the RRTIM in the program, this should start in year one	12	20
Offer more guidance from the teacher, such as: (1) more feedback on the issues that arise in the student teams (2) more guidance in the roles, from teachers that are trained in the roles, and ask students about the tasks that come with their role	5	3
Change aspects of the RRTIM, such as: (1) provide more concrete assignments/tasks, also linked with professional practice (2) more check-ins (3) make a more clear link between the roles and the research project, as well as professional roles (4) make it more clear which task belongs to which role, and make the roles less complicated (5) rather discuss issues than assign roles (6) rather use an existing technique such as scrum (7) change the role of the communication manager as it has too much overlap with other roles (8) add more lectures on collaboration (9) make sure the roles do not create extra workload	15	11
Change aspects of the lecture	10	10
Address encountered problems in the execution of the roles, such as: (1) the division of roles does not stay intact during the project, students take on each other's roles	8	10

(Continued)

TABLE 2 Continued

Theme	unique comments	students
(2) it is difficult to make decisions as manager, would be good to be trained in that earlier in the program		
(3) it works if everyone is taking on their role, otherwise it can lead to taking less responsibility for the project		
(4) lack of physical space to collaborate on campus		
(5) it is difficult to hold others responsible for executing the tasks assigned to their role, if they do not take up the responsibility		

supervising and coaching student teams, 3) the introductory lecture in which the roles were explained was placed at the beginning of the course and more attention was paid to the importance of learning teamwork and the competence of reflection, 4) the manuals were rewritten from a how-to-fix-problem-X structure to a thematic structure and updated based on students', teachers' and experts' feedback, 5) more clarification and guidance per role was offered, 6) more scaffolding was provided for the reflective team conversation on the (sub-)optimal team processes in the team, 7) the RRTIM was mandatory for all students and part of the course outline.

Validation of the RRTIM PART 3: evaluation of the RRTIM

After the RRTIM was redesigned, it was implemented in a first year, second semester course in program 3 (course 2). In another course (course 1), only minimal scaffolding of teamwork was provided, which consisted of drawing up a team charter, reflection thereon and a final reflection on teamwork. The idea was to examine whether students' teamwork competence and level of reflection on teamwork differed between minimal scaffolding and the full RRTIM, based on three indicators: (a) whether their level of reflection on teamwork increased, (b) whether they reflected on more teamwork aspects, (c) the number of activities they reported undertaking to improve their own teamwork.

Students in both course 1 and 2 showed a quite shallow level of reflection, but students in course 2 did reflect slightly more in-depth in course 2. For example, student 41 did not include their own frame of reference and wrote sentences like 'giving feedback went well in our collaboration' in course 1, and in course 2 the same student did include their own frame of reference and wrote sentences like:

Outside of class, we collaborated by video calling. I personally found this difficult, because I prefer meeting and collaborating in person. Then I work faster, more accurately and of course I find it more socially enjoyable. I really had to adapt to the rest of the group, because they felt about it differently.

Of course, it cannot be ruled out that there was also a maturation effect as students had progressed a few months in their study programme.

Students also reflected on more teamwork aspects in course 2 than in course 1. They reflected both in a positive sense (this aspect went well) as in a negative sense (this aspect could have been improved), with a total of 42 aspects mentioned in course

1 and a total of 55 aspects mentioned in course 2. Only five students reflected on less teamwork aspects in course 2 than in course 1. Some teamwork aspects received much more attention from students than others: communication (mentioned by 72 students), planning (44), affect (43), task division (42), back-up and helping (32), atmosphere (28), making and keeping agreements (26), openness (24), balance in the team (19), feedback (19), availability of team members (18), commitment/contribution of team members (18), motivation (17), and shared mental models (17) were the most mentioned teamwork aspects. For almost all students, the same student reflected mostly on different teamwork aspects in course 1 than in course 2, underscoring the dynamic and emergent nature of teamwork. Teamwork and team dynamics are different in every team. Most students only repeated 0–3 aspects, which is consistent with the idea that students adapted their reflection to the specific dynamics in the team and the aspects that needed attention within that particular team in both courses.

The most striking difference between course 1 and course 2 is that students reported much more activities they undertook to improve their teamwork, both as individuals and as a team. In course 1 students, reported a total of 37 activities they undertook to improve their own teamwork, of which 17 were undertaken as an individual (e.g., 'listen to others actively and ask them questions') and 20 as a team (e.g., 'organize more meetings'). However, students also reported explicitly that they had encountered a problem and had done nothing (four students) or that they wished that had undertaken an activity but did not (18 students, 25 comments). Some examples of activities they wished they had undertaken were 'discuss the task division earlier and better', 'plan more face-to-face meetings', 'ask for help', and 'confront a team member with their behavior'.

In course 2, the same students reported considerably more activities (a total of 165 activities) they undertook to improve their teamwork, of which 96 as an individual (e.g., 'trying to find a middle ground when conflicts arise between team members') and 69 as a team (e.g., 'discuss issues within the team'). Students still voiced 22 things they would have liked to do differently next time, although these often concerned applying a lesson learned later in the course at an earlier stage of their teamwork.

The individual activities to improve teamwork were often in line with the student's role, and the team activities were partly initiated by students in connection to their role or the mandatory reflective team conversation they had during the course. As respondent 39 describes:

What worked well was that I helped the team to lay out different perspectives side by side when we faced complex decisions. For example, when we were hesitating between two directions for our prototype, I actively put all arguments and assumptions on the table and worked with the group to look for common ground. This enabled us to reach a strong decision. What went less well was that, in the beginning, I sometimes struggled to step in on time when we were drifting off too much. As a result, the transition from brainstorming (diverging) to making decisions (converging) was sometimes too slow. This occasionally created tension, because people became frustrated with the pace. We solved this by agreeing on clearer phases and making decisions at fixed moments. This worked well, but in hindsight I should have taken the lead on this sooner.

This quote shows the interplay between the student's analysis of a sub-optimal team process based on observations from their role ('the transition from brainstorming to making decisions was too slow'), individual action based on the role ('laying out different perspectives' as part of the task manager role), team action as a result of the reflective team conversation ('agreeing on clearer phases and making decisions at fixed moments'), and a continued reflection on the student's own action ('in hindsight I should have taken the lead on this sooner'). These type of reflections were much more frequent in course 2 than in course 1. Another example of how students actively intervened in sub-optimal team processes is illustrated in the following reflection by respondent 25:

What went less well was that, because of the positive atmosphere, it turned out to be more difficult to openly give criticism. During [reflective] team meeting 3, we realized that all of us had sometimes held back our criticism. To improve this, we added a section to our shared document where we could express doubts and critical points. In fact, that conversation already helped us start a new conversation in which we shared our doubts more. This helped us refine our product.

Overall this exploratory comparison suggests that in the full RRTIM course, students showed more awareness of different aspects of teamwork, slightly more in-depth and nuanced reflections upon and analysis of their own teamwork, and more reported agency in changing sub-optimal team processes and strengthening already good team processes.

Discussion and conclusion

In this paper we describe the development of the RRTIM, which was designed to scaffold students' learning of teamwork competences and to help students learn to navigate dynamic and emergent team dynamics during interdisciplinary teamwork. In higher education programs teamwork competences are often not explicitly developed (Hansen, 2006), but students do ask for explicit teamwork teaching (van Woerden et al., 2025a). The first version of the RRTIM was received quite well by students and teachers, but needed further improvement. The main

improvements were to further embed the RRTIM in the program and course (also by making it a mandatory part of a course), to train and guide teachers and students in the RRTIM, to shorten and restructure the role manuals, and to provide more scaffolding for the reflective team conversation on (sub-) optimal team processes. In our explorative comparison of a minimal scaffolding to a full RRTIM course, analysis showed that in the full RRTIM course, students showed slightly more in-depth and nuanced reflection on teamwork, more awareness of a wider range of different teamwork aspects and more reported agency in changing sub-optimal team processes and strengthening already good team processes. The role-based method seems to help students in paying attention to a wider range of teamwork aspects while at the same time offering them more agency to change sub-optimal team processes.

Some issues however remain to be solved. First of all, the RRTIM seems to work especially well for teams that are considered 'good' teams by the teachers. Student teams that function less well, have more difficulty to make use of the RRTIM and to have good reflective team conversations on their team processes and team dynamics. In less functioning teams, there are more sub-optimal team processes to discuss, which may increase students' fear of individual, social, and academic consequences of addressing these issues (van Woerden et al., 2025a). Also, in some of these teams there seems to be a lack of buy-in in the reflective activities, which is in line with findings of Lescott (Lescott, 2022). This seems to be the main issue in scaffolding student teamwork competence learning: how to help students have fruitful conversations about what can be improved in their teamwork, especially in teams that function less well and need to have these conversations more urgently. In the last version of the RRTIM, students had to report on their team conversations as a mandatory (but ungraded) assignment, which did improve their engagement with the reflection on their (sub-) optimal team processes. However, even when students engage in the reflection activity, they seem to either keep their discussion superficial or steer it in a different direction when the level of tension or fear of individual, social or academic consequences becomes too high. So even though reflection is theorized to be a powerful tool for learning (Kolb, 1984; Moon, 2004; Schön, 1987), students can refrain from that learning if other potential costs are perceived as too high.

Another issue with further implementation of the RRTIM is that not all university teachers feel equipped or feel it is part of their job to guide teamwork processes. As Schrujer (2016) notes a teacher needs to have the skill and the courage to manage team dynamics, of which the teacher is also a part. In line with findings by Riebe et al. (2016) educators are not always able, competent or willing to guide teamwork processes. Both students and teachers need support in the teaching and learning of teamwork competences and the RRTIM was redesigned to offer more of that support. However, even though the RRTIM includes a training to enhance teachers' ability to guide teamwork processes and is offering ready-to-use materials (such as the manuals for students), their attitudes towards it may be unchanged. If teachers do not wholeheartedly integrate the RRTIM into the course, then students that are already reluctant to use the method may have even less buy-in.

Limitations

The research in this paper was done in one university in the Netherlands, in three interdisciplinary programs. There may be cultural and/or educational differences between universities and countries, that may render different research results in other contexts. Also, the outcomes of the RRTIM was only exploratorily researched, and more empirical testing of the method is needed in the same as well as in other contexts. In addition, more longitudinal data could be collected to understand the long-term effects. Thirdly, for the analysis we had to rely on what students reported in their reflections, so some of the team dynamics or activities they undertook may not have been reported and therefore not represented in the analysis. Finally, as the students are compared between two courses, it is possible that some of the reported positive effects of the RRTIM may not be attributed to the implementation of the method, but to a maturation effect.

Implications

We show that a role-based method of guiding students' teamwork can help students identify a broader range of teamwork aspects at play in their teamwork, and show more agency in changing sub-optimal team processes. Even if students do not always stick to the roles, the method seems to help them become aware that teamwork consists of many interrelated aspects, that all need their attention. Students do not only identify (sub-)optimal team processes more easily in the course where the full RRTIM was implemented, they also undertake more action to change these processes, both individually and as a team, and reflect on the effectiveness of these changes. These findings suggest that students more fully complete Kolb's (1984) reflective cycle in the RRTIM course.

Where Horn et al. (2022) offer empirical support for how students' reflection on teamwork experiences of fictional characters improves their interdisciplinary consciousness and teamwork competences, we offer empirical support that students' reflection on their *own* teamwork may also improve their teamwork competence. In addition, students had a higher level of reflection in the RRTIM course. However, as their own teamwork is more high-stakes than fictional teamwork, they also experience more challenges in the reflection activity, which needs adequate guiding by teachers.

For educators in interdisciplinary higher education, the RRTIM may prove to be a useful method to scaffold student learning of teamwork competences and navigating complex, dynamic and emergent team dynamics. Students have voiced the need for more scaffolding of teamwork competence learning (van Woerden et al., 2025a), and using the RRTIM they show higher levels of reflection on teamwork, a wider range of teamwork aspects that they pay attention to and more agency in changing sub-optimal teamwork processes.

Data availability statement

The datasets presented in this article are not readily available because the ethical advisory board has requested not to share

student data with others. Requests to access the datasets should be directed to r.vanwoerden@uva.nl.

Ethics statement

Ethical approval for Part 1 was not required for the study involving humans in accordance with the local legislation and institutional requirements. The studies involving humans in Part 2 and 3 were approved by the Ethical Advisory Board of the Amsterdam Institute for Social Science Research. The studies were conducted in accordance with the local legislation and institutional requirements. The participants provided their written or verbal informed consent to participate in this study.

Author contributions

RV: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing. AW: Conceptualization, Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing. MH: Conceptualization, Funding acquisition, Investigation, Project administration, Resources, Writing – original draft, Writing – review & editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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Supplementary material

The Supplementary Material for this article can be found online at: <https://www.frontiersin.org/articles/10.3389/feduc.2026.1840604/full#supplementary-material>

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